

Two Final Submissions, One System

Technical description of submissions 022 and 026 of team MiyakoHAR, CUHK-X 2026 Multimodal Human Activity Challenge, Small Model Track

Joseph Arthur Koo* Ken Argani Toendan Nhung Huyen Hoang Zilu Liang (faculty advisor)

Ubiquitous and Personal Computing Lab, Kyoto University of Advanced Science, Kyoto, Japan

*Corresponding author: 2023m620@kuas.ac.jp 14 September 2026

Summary

Team MiyakoHAR selected two final submissions for the CUHK-X 2026 Small Model Track: submission 022 (022_r152_fence.csv, public score 0.76616, 154 of 201 clips) and submission 026 (026_hr320_person_pixels.csv, public score 0.78606, 158 of 201). Both come from the same code, the same training data and the same two-branch design. Each is one file of 83.5 MB (022) or 83.6 MB (026) that holds two 3D convolutional networks: an ir-CSN-R50 that reads the depth and infrared streams together, and an ir-CSN-152 that reads the thermal stream. The two networks' class probabilities are averaged, and the largest entry is the predicted activity. Submission 026 differs from 022 in one network only. Its thermal network was retrained on thermal frames decoded at their native 320×240 size instead of a reduced 160×120 copy, and it reads a 320×320 input instead of 224×224 . Everything else, including the depth and infrared network, is byte for byte the same.

Whichever of the two the private leaderboard scores higher is the artefact we reproduce at verification. For both, a fresh clone of our repository plus that one file reproduces our submitted CSV byte for byte on a CPU through the same entry script the committee will run: 14 minutes end to end for 022 and 23 minutes for 026. Inference is per clip and stateless, runs offline, loads nothing but the one file, and uses no pseudo-labels, no test labels and no data beyond the competition's own training split.

One request for the verification sample: please read the seen subjects (Part A) and the unseen subjects (Part B) separately, and per subject where possible. On our own leave-subjects-out instrument the same model spans 62% to 84% across unseen subjects.

1 Purpose and scope

The competition rules state that a team “may be required to provide a detailed, reproducible description of how the winning Submission was generated (architecture, preprocessing, training details, hyper-parameters, etc.), together with a link to a code repository with complete instructions”. This document is that description, written for both of our final submissions. It is addressed to the organizers and the verification committee first and to our own team second, so every specialist term is defined where it first appears.

We describe two submissions rather than one because both are selected as finals and the private leaderboard decides which of them counts and which we reproduce at verification. The two share every component except one trained network, so Sections 3 to 5 describe the shared design once and state the single difference where it applies. Section 6 gives the reproduction procedure, which is the same for both apart from the file placed in the checkpoint slot and the CSV it is compared against.

This document does not report a validation result as a leaderboard gain, and it does not describe the 25 earlier submissions except where one explains a design choice. The complete experiment

record, a dated ledger of every decision and measurement, is kept in the team’s planning repository and is available to the committee on request.

2 The task and the data

A Task

CUHK-X 2026 is a cross-subject activity recognition challenge: 40 classes of activities of daily living, recorded at home in several sensing modalities, scored by top-1 accuracy against a single label per clip. The Small Model Track limits everything loaded at inference to one checkpoint of at most 100,000,000 B, and the organizers confirmed that the limit covers every model used at inference rather than the recognition model alone. Inference runs from the original test files and without network access. Eighteen subjects appear in the training split. Four other subjects appear only in the test split, which holds 405 clips scored as 201 public and 204 private. A model therefore has to recognize activities performed by people it has never seen.

B Streams

Table 1 lists the released streams and what our submissions read. Depth and infrared are pixel-registered at a common origin, so one network reads them as a four-channel stack. Thermal comes from a different sensor: it covers 59% of the infrared field of view and its frames carry a counter rather than a wall-clock time, so aligning it frame by frame with the other two would carry more than 0.5 s of error on 20 to 23% of clips. For that reason thermal has a network of its own, and the two networks are combined at the probability level.

Table 1. The released streams and their use in both submissions.

Stream	Released as	Used	Read by
Depth colormap	640 × 480, 3 channels	yes	the depth + infrared network, channels 1 to 3
Infrared	640 × 480, 1 channel	yes	the depth + infrared network, channel 4
Thermal	320 × 240, 3 channels	yes	the thermal network
IMU, radar, skeleton	time series and key points	no	not read by either submission

C Training data

The training split holds 2,931 clips with depth and infrared frames and 2,891 clips with thermal frames, all from the 18 training subjects (users 1 to 9 and 16 to 24). No other data was used to train either submission, and no test clip, test label or pseudo-label enters either of them. The label of every training clip comes from its directory path and from nothing else. Of the 405 test clips, 4 carry no infrared frames and 10 carry no thermal frames; none is missing both.

3 The recognition system

A Two branches, one probability vector

Both submissions have the shape shown in Figure 1. A clip arrives as up to three raw streams and leaves as one of 40 labels. First, the depth colormap (3 channels) and the infrared frame (1 channel) are stacked into a four-channel input and read by an ir-CSN-R50, a 3D convolutional network with channel-separated convolutions [1]. Second, the thermal frame is read by an ir-CSN-152, the deeper network of the same family. Third, each network produces a vector of 40 softmax probabilities, the two vectors are averaged over the branches present in the clip, and the largest

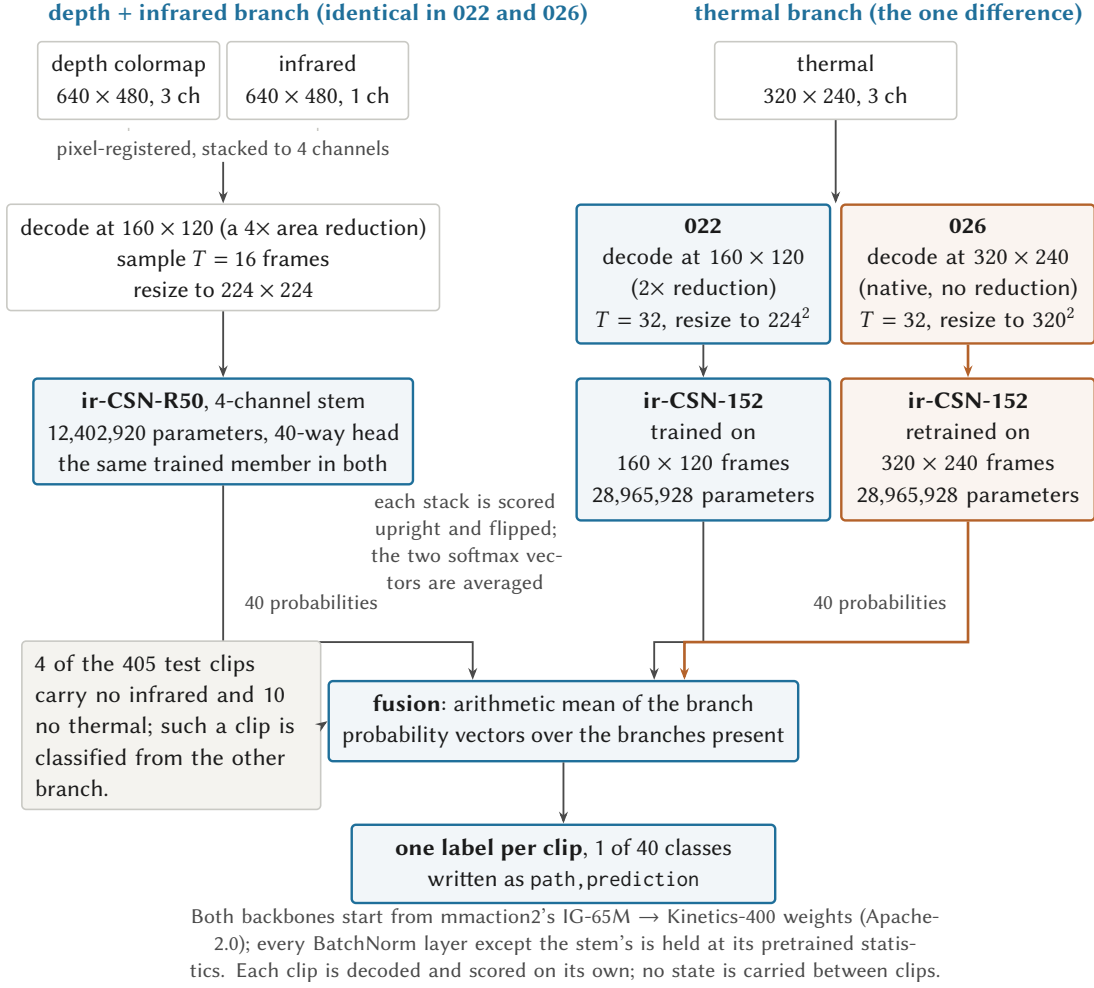


Figure 1. The recognition system of both final submissions. Depth and infrared are pixel-registered and travel together as a four-channel stack into an ir-CSN-R50; thermal comes from a separate sensor and travels alone into an ir-CSN-152. Each network produces 40 class probabilities, the two vectors are averaged, and the largest entry is the label. The two submissions share the left branch byte for byte. They differ only in the thermal branch: in 022 the thermal frames are reduced to 160×120 and read at 224×224 , in 026 they are decoded at the sensor's native 320×240 and read at 320×320 by a network retrained for that input.

entry is the prediction. A clip that lacks one modality is classified from the other branch alone rather than dropped.

Both networks start from published weights, mmaction2's ir-CSN checkpoints pretrained on IG-65M and fine-tuned on Kinetics-400 (Apache-2.0) [2, 3, 4], and are then fine-tuned on the competition's training split with 40-way classification heads. The ir-CSN family was chosen after the pretraining corpus proved to be the largest single effect we measured: replacing the earlier Kinetics-only backbone with ir-CSN-R50 moved our fold instrument by +5.075 pp. The deeper ir-CSN-152 ships only on the thermal side, where the smaller network was capacity-limited; on the depth and infrared side the deeper network did not pay. During fine-tuning every BatchNorm layer except the stem's is held at its pretrained statistics (52 of the 53 in the R50), because those statistics are what the pretraining exists to supply. The depth and infrared stem is widened from 3 to 4 input channels with a response-preserving $\times 0.75$ rescaling of the pretrained filters.

B What differs between 022 and 026

Table 2 places the two submissions side by side. The depth and infrared network is the same trained member in both, byte for byte. The thermal network differs: the member in 022 was trained on thermal frames decoded at 160×120 and reads a 224×224 input; the member in 026 was trained on frames decoded at the sensor’s native 320×240 and reads a 320×320 input, at a training batch of 6 instead of 12. Nothing else in the recipe changed. The two architectures, and therefore the parameter counts, are identical: 12,402,920 parameters in the depth and infrared network and 28,965,928 in the thermal network, 41,368,848 in all, counted on the packaged file with the 40-way heads. The two files differ in size by 16,546 B, which we attribute to the stored metadata described in Section 4, because the weights have identical shapes. On the 405 test clips the two files disagree on 42 predictions; both predict all 40 classes (Appendix A).

Table 2. The two final submissions side by side.

	Submission 022	Submission 026
Submitted CSV	022_r152_fence.csv	026_hr320_person_pixels.csv
Submitted on (JST)	29 August 2026	13 September 2026
Public score	0.76616 (154 of 201)	0.78606 (158 of 201)
Artefact file	one fp16 file, 83,543,089 B	one fp16 file, 83,559,635 B
SHA-256 (first 16 hex)	218266cc3230e87f	cd27bc72c09f6e41
Depth + infrared network	ir-CSN-R50, $T = 16$, 224^2 input	the same member, same bytes
Thermal network	ir-CSN-152, $T = 32$, 224^2 input	ir-CSN-152, $T = 32$, 320^2 input
Thermal frames decoded at	160×120 (a $2\times$ reduction)	320×240 (native)
Thermal training batch	12	6
Parameters (both networks)	41,368,848	41,368,848
Predictions that differ	42 of 405 clips	
CPU reproduction, fresh clone	855 s end to end	1,379 s end to end

C What is not in the system

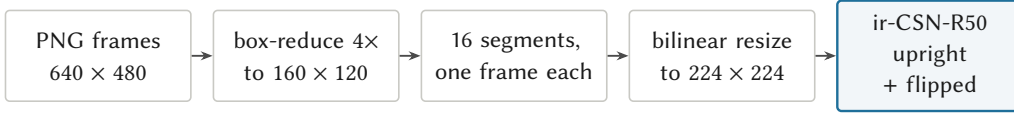
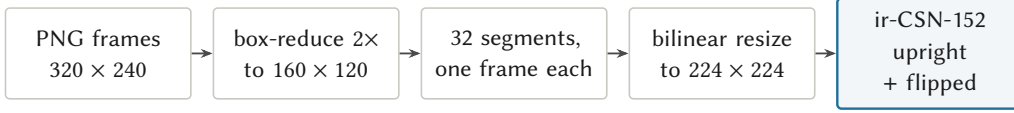
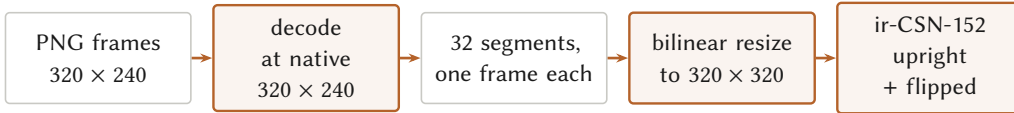
No person detector and no learned preprocessing model is loaded: the shipped path decodes the full frame of every stream and passes it to the networks. No language model and no external service is called, no adaptation happens at test time, and no state is carried from one clip to the next. The repository also contains tools and members from earlier experiments, among them a pseudo-labelled submission (023) and a ViT-B container (025); none of them is part of either final, and Section 9 states why.

4 Preprocessing at inference

Everything in this section is executed by `inference.sh` from the original test files. There is no separate preprocessing step for the committee to run, and no preprocessing artefact is shipped. Figure 2 draws the path of one clip.

A Decoding

Each clip is a directory of PNG frames per stream. Depth colormap and infrared frames (640×480) are decoded and reduced $4\times$ to 160×120 by box (area) averaging. Thermal frames (320×240) are reduced $2\times$ to 160×120 in submission 022, and are decoded at their native 320×240 with no reduction in submission 026. The 026 thermal member carries a stamped key in its checkpoint metadata, `source_wh = [320, 240]`, which the loader reads and applies to that member’s stream. A member without the key is decoded exactly as before, bit for bit; this is why the shared depth

depth + infrared stream, both submissions**thermal stream, submission 022****thermal stream, submission 026 (the stamped key source_wh = [320, 240] selects this row)**

Every resize keeps the trainer's geometry: a 4:3 frame is stretched 1.33× vertically to a square, the same stretch every member learned on. After the backbones, the two views of a stack are averaged, then the branches, then the largest probability is written. No crop, no detector and no learned preprocessing is applied in either submission.

Figure 2. The path of one clip through inference.sh, read left to right for each stream. The depth and infrared row is identical in both submissions. The two thermal rows differ in the first two boxes only: submission 022 reduces the 320×240 thermal frames to 160×120 before sampling and resizes them to 224×224 , while submission 026 decodes them at their native size and resizes to 320×320 for a network retrained on that input. The choice is made by a key stored in the 026 checkpoint, so the same code reproduces either submission from its own file.

and infrared member behaves identically in both submissions, and it was proved by reproducing the 022 CSV byte for byte under the new code.

B Sampling

Frames are sampled by temporal segment sampling [5]: the clip is divided into T equal segments and one frame is taken from each, at a fixed position at inference so the result is deterministic. $T = 16$ for the depth and infrared network and $T = 32$ for the thermal network; each member stores its own T in the checkpoint and the loader builds the clip accordingly.

C Resizing

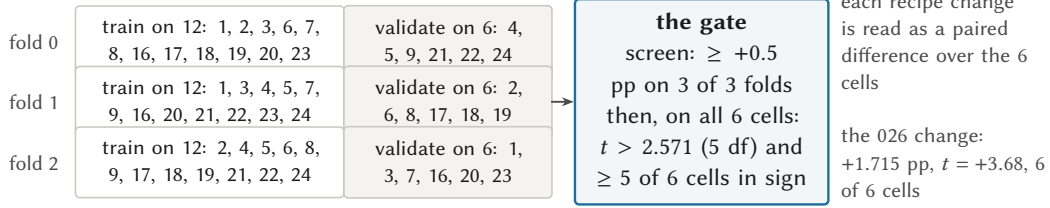
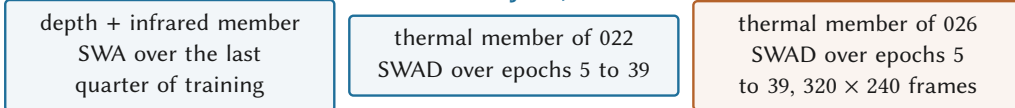
The sampled 160×120 frames are resized bilinearly to 224×224 . The aspect ratio is not preserved: the frame is stretched 1.33× vertically, and every trained member learned on exactly this geometry. The 026 thermal member resizes its 320×240 frames to 320×320 with the same stretch.

D Test-time flip and fusion

Each frame stack is scored twice, upright and horizontally flipped, and the two softmax vectors are averaged. The branch vectors are then averaged over the branches present in the clip (an arithmetic mean), and the class with the largest probability is written to the CSV.

E Damaged or missing input

A clip that lacks one stream is classified from the other branch. The entry point does not stop on a missing, short, corrupt or odd-resolution clip: the robustness test runs the real entry point on six such cases and requires a written prediction for each (6 of 6 pass for both artefacts). The

18 training subjects (users 1 to 9 and 16 to 24)**A. the selection instrument: leave 6 subjects out, 3 folds, 2 seeds****B. the shipped members: trained on all 18 subjects, no validation set**

Because no subject is held out, the shipped weights are chosen by a weight average over the run, not by a best validation epoch. The instrument in A decides which recipe to ship; the members in B are what ships.

Figure 3. How the training data was used. Part A is the measuring instrument: a frozen split holds out six subjects per fold and trains on the other twelve, so every subject is validated once by members that never saw it; with two seeds a recipe change is read as a paired difference over six subject-disjoint cells and must pass a screen and a gate. Part B is what ships: once a recipe passed, each member was retrained on all 18 subjects with a weight average as the checkpoint selector, so no validation clip selects the shipped weights.

output is a CSV with the header path, prediction and one integer class per test clip, 405 rows.

5 Training

A Data used

Every shipped member was trained on the clips of all 18 training subjects and on nothing else. Because no subject is held out, there is no validation set, and the shipped weights are selected by a weight average over the training run rather than by the best validation epoch, so no validation clip selects them. Figure 3 separates the two uses of the data: the leave-subjects-out instrument that decided which recipe to ship (Section D), and the final training on all 18 subjects that produced the shipped members.

B The recipe

Table 3 lists every hyper-parameter of the three trained members. The optimizer is SGD with momentum 0.9 and Nesterov acceleration at a learning rate of 5×10^{-4} , with a 3-epoch linear warmup counted in iterations, weight decay 5×10^{-4} , label smoothing 0.1, gradient-norm clipping at 20, and 40 epochs. The learning rate 5×10^{-4} is the measured optimum for this backbone family; the inherited 5×10^{-3} destabilized it on 6 of 6 cells. Batch 12 was measured as worth +2.116 pp over the inherited default of 16 on this family. The depth and infrared member follows a cosine schedule to zero and keeps a stochastic weight average [6] over the last quarter of its iterations. The two thermal members keep a constant learning rate after warmup and average the weights densely, at every iteration, over epochs 5 to 39 (SWAD [7]); they also use a temporal crop augmentation.

C The change from 022 to 026, and why

The person occupies about 5.9% of a thermal frame, so at 160×120 the classes that depend on the hands and on hand-held objects are decided on a few dozen source pixels. Decoding at 320×240

Table 3. Hyper-parameters of the three trained members. The first two members form submission 022; the first and the third form submission 026.

	Depth + infrared member	Thermal member (022)	Thermal member (026)
Run name	depthir_all118_alien_q16_ircsn224_s1	thermal_all118_labpc_r152stack_s1	labpc_hr320_thermal_all118_r152stack_s1
Architecture	ir-CSN-R50, 4-channel stem	ir-CSN-152	ir-CSN-152
Initialization	IG-65M $\rightarrow$ Kinetics-400	IG-65M $\rightarrow$ Kinetics-400	IG-65M $\rightarrow$ Kinetics-400
Source frames decoded at	160×120	160×120	320×240
Input size	224×224	224×224	320×320
Frames per clip T	16	32	32
Batch size	12	12	6
Optimizer	SGD, momentum 0.9, Nesterov	same	same
Learning rate	5×10^{-4}	5×10^{-4}	5×10^{-4}
Schedule	3-epoch warmup, cosine to 0	3-epoch warmup, then constant	3-epoch warmup, then constant
Weight averaging	SWA, last quarter of iterations	SWAD, epochs 5 to 39	SWAD, epochs 5 to 39
Epochs	40	40	40
Weight decay	5×10^{-4}	5×10^{-4}	5×10^{-4}
Label smoothing	0.1	0.1	0.1
Gradient-norm clip	20	20	20
Temporal crop augmentation	no	yes	yes
BatchNorm	held except the stem	held except the stem	held except the stem
Training subjects	all 18	all 18	all 18
Seed	20260813	20260813	20260813
Machine	Alienware workstation	RTX 3090 workstation	RTX 3090 workstation

gives the thermal network every pixel the sensor recorded. The batch went from 12 to 6 because the ir-CSN-152 at a 320×320 input needs 18.3 GB of the 24 GB card at batch 6; the fit check before the first training cell measured 18.339 GB. This was the thirteenth of fourteen pre-registered experiments that could change the shipped model, and it was the first whose fused result cleared our promotion gate on an instrument with no exposure to test data.

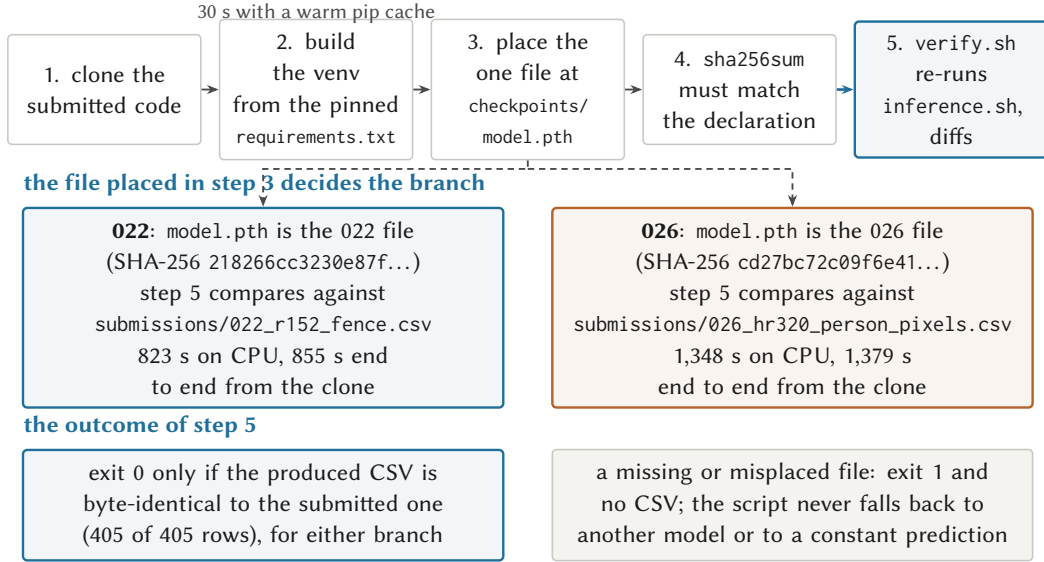
D How the recipe was selected: the fold instrument

Every recipe change was judged on a frozen 3-fold subject-grouped split of the 18 training subjects, generated once from a recorded seed and never changed (Figure 3, part A). Each fold trains on 12 subjects and validates on the other 6, so every subject is validated exactly once by members that never saw it. With two training seeds this gives six subject-disjoint cells per recipe. A change was adopted only if the fused two-branch model improved by at least +0.5 pp on all three folds at a first screen and then, over all six cells, passed $t > 2.571$ on 5 degrees of freedom with at least 5 of the 6 cells in the direction of the effect. The number of such gated reads was capped in advance.

Table 4 gives the six cells for the two recipes. The 026 recipe improves on the 022 recipe by +1.715 pp on average (sd 1.141, $t = +3.68$, 6 of 6 cells). Fold 1 is the weak fold at both seeds. This is a validation reading. On the public leaderboard, 026 then scored four clips above 022 (158 against 154 of 201), inside the range of 152 to 166 that we wrote into the record before the score existed. A difference of four clips on 201 is inside one standard error of a public score, which is six clips at this accuracy, so we describe 026 as consistent with its fold result and not as a measured gain.

Table 4. Fused two-branch accuracy on the six leave-6-subjects-out cells, in percent. The 022 recipe decodes thermal at 160×120 for a 224^2 input; the 026 recipe decodes it at 320×240 for a 320^2 input.

Recipe	Seed 1			Seed 2			mean
	fold 0	fold 1	fold 2	fold 0	fold 1	fold 2	
022 recipe	73.975	72.693	71.834	74.898	72.227	70.719	72.724
026 recipe	76.025	73.253	75.076	76.332	72.600	73.354	74.440
Difference (pp)	+2.049	+0.559	+3.242	+1.434	+0.373	+2.634	+1.715



On the committee's own sample data the same entry point runs with their row list (`./inference.sh <data_dir> <out.csv>`), or `-test-csv <their_list.csv>` for arbitrary clip names). A GPU makes step 5 about twelve times faster and changes no prediction.

Figure 4. The reproduction procedure, in the five commands of the README. Steps 1 to 4 are identical for both submissions; step 3 places the one file that the private leaderboard selected, and step 5 compares the re-run with that submission's CSV. Every timing shown was rehearsed on the verification machine from a fresh clone with a cold environment. The script exits 0 only on a byte-identical file, and it refuses to run rather than fall back when the artefact is missing.

E Compute

The six fold cells of the 026 thermal recipe cost 15.2 GPU-hours and the final member trained on all 18 subjects cost 3.0 hours, all on one RTX 3090 with 24 GB. The depth and infrared member was trained on a separate workstation; the two thermal members on the RTX 3090.

6 Reproduction and verification

Figure 4 shows the procedure the committee follows. It is the same for both submissions; only the file placed in the checkpoint slot and the CSV it is compared against change.

A Environment

The pinned interpreter environment is `requirements.txt`: `torch==2.13.0`, `torchvision==0.28.0`, `numpy==2.4.4` and `Pillow==12.2.0` on CPython 3.12. The inference path imports nothing else, and that is asserted by execution: the robustness test runs the real entry point with a poisoned `yaml` module on the path. `PyYAML==6.0.3` is needed only by `train.sh`. A GPU is optional. CPU-only wheels reproduce every accuracy number; only the timings differ. Every number in this

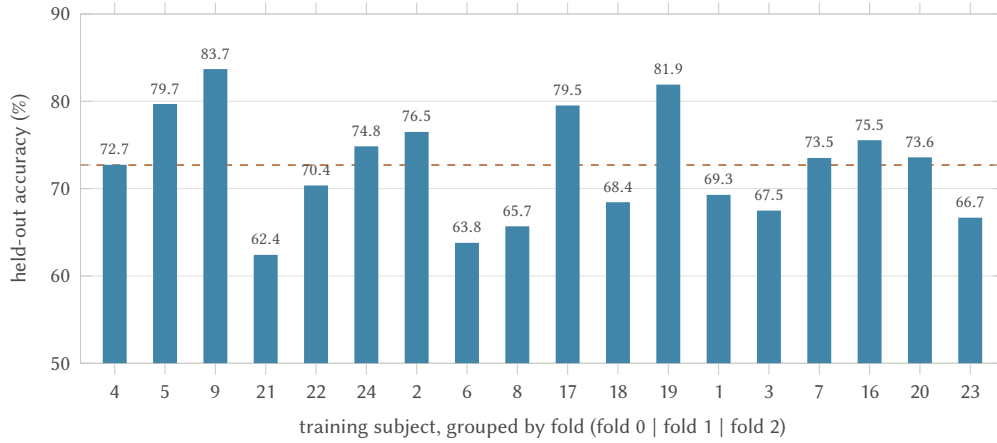


Figure 5. Accuracy of the 022 recipe on each training subject when that subject is held out, averaged over two seeds. The same model spans 62.41% (subject 21) to 83.69% (subject 9), a between-subject standard deviation of 6.2 pp around a clip-weighted mean of 72.7% (the dashed line). The spread is a property of subjects, not of the sample size: subjects 21 and 6 sit more than 7.3 pp below the mean on their own. This is why we ask that the verification sample be read per subject and that seen and unseen subjects be reported separately.

document was verified on one machine: Ubuntu 24.04, an RTX 3090 with 24 GB, NVIDIA driver 595.84, CPython 3.12.3; its full pip freeze is shipped as `environment-labpc.txt`. The depth and infrared member was trained on CPython 3.12.13, a patch-level difference we record rather than hide. If `python3 -m venv` stops because `ensurepip` is missing, any of `apt install python3-venv`, `virtualenv .venv` or `uv venv .venv` works, and `inference.sh` prints the same three ways when it refuses to run without torch.

B The five commands

```
git clone <the submitted code> repo && cd repo
python3 -m venv .venv && \
.venv/bin/pip install -r requirements.txt
mkdir -p checkpoints && \
cp <upload>/checkpoints/model.pth checkpoints/model.pth
sha256sum checkpoints/model.pth
./verify.sh <data_dir> submissions/022_r152_fence.csv
# for submission 026: ./verify.sh <data_dir> submissions/026_hr320_person_pixels.csv
```

`<data_dir>` is the directory that contains `small_model_track_test/`. The SHA-256 printed by the fourth command must equal the value declared for the artefact being reproduced (Table 7). The fifth command is the reproduction itself.

C What `verify.sh` checks

The script prints the SHA-256 of `checkpoints/model.pth`, weighs the file against the 100,000,000 B limit and against our own 95 MB gate, re-runs `./inference.sh` on the test clips, and compares the produced CSV with the submitted one. It exits 0 only when the two files are byte-identical. If the artefact is missing or at another path, it exits 1 and writes no CSV; it never falls back to a different model or to a constant prediction. Table 5 lists the rehearsed timings. Both artefacts were reproduced from a fresh clone with a cold virtual environment on the CPU, 405 of 405 rows identical.

Table 5. Rehearsed reproduction and inference timings on the verification machine.

	Submission 022	Submission 026
verify.sh on CPU, fresh clone, cold environment	823 s	1,348 s
End to end including clone and environment build	855 s	1,379 s
CPU time per clip, 8 threads	1.91 s	3.45 s
GPU time per clip, RTX 3090	0.159 s	0.221 s
GPU wall time for 405 clips	65 s	about 90 s
Peak GPU memory, RTX 3090	908 MiB allocated	1,679 MiB allocated
Peak CPU memory	1.66 GB	2.43 GB
Organizers' ceiling for the live run	120 minutes	

D Statelessness and offline operation

Two properties matter for a verification run on new data, and both are tested rather than asserted. First, inference is per clip: a random subset of 40 test clips reproduces the full run's predictions for those clips exactly, 40 of 40, for both artefacts (131 s on CPU for 022, 138 s for 026). The result of a clip therefore does not depend on which other clips are in the batch. Second, inference needs no network: the entry point is run inside a fresh network namespace with the loopback interface down and empty model caches, and it must write the submission (pass, both artefacts).

E Running on the committee's sample data

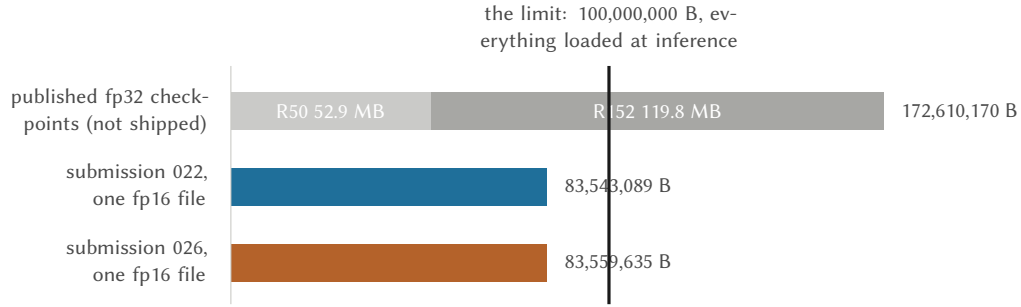
The row list of the output CSV comes from the organizers' CSV whenever one exists, and only otherwise from directories named `SM_test_<digits>`. On sample data with other clip names, either pass the list explicitly, `python src/predict.py <data_dir> -test-csv <their_list.csv> -out <out.csv>`, or copy it to `<data_dir>/test.csv` and run `./inference.sh <data_dir> <out.csv>` unchanged. Without a row list and without `SM_test_*` names the entry point stops with a message rather than guessing. This was rehearsed on a renamed copy of five test clips, and both CSV routes reproduced the shipped predictions clip for clip.

Figure 5 shows why we ask that the sample be read per subject. When each training subject is held out in turn, the 022 recipe scores between 62.41% (subject 21) and 83.69% (subject 9) on the held-out subject, with a clip-weighted mean of 72.7% and a between-subject standard deviation of 6.2 pp. Two of the 18 subjects sit more than 7.3 pp below the mean on their own. A sample mean over two unseen subjects has a standard deviation of about 4.2 pp and differs from the population mean by more than 7.3 pp in about 8% of draws, by sampling alone. The seen half of the sample reads differently: every shipped member was trained on all 18 training subjects, so clips of those subjects are in-sample and will score above the held-out numbers here. We ask the committee to read Part A and Part B separately, and per subject where it can.

7 Size and provenance of every weight

Figure 6 draws the byte budget to scale. The organizers' limit of 100,000,000 B covers the total size of all models used at inference, and each of our artefacts is one file that holds both trained networks in half precision: 83,543,089 B for 022 and 83,559,635 B for 026, each weighed on disk by three agreeing counts. The published checkpoints the networks start from are 52,855,745 B (ir-CSN-R50) and 119,754,425 B (ir-CSN-152) in single precision with their 400-way Kinetics heads; the shipped file replaces those heads with 40-way heads and stores every tensor in fp16. The file carries 41,368,848 parameters, counted on the packaged file itself.

Table 6 names every pretrained weight that enters either artefact, with its source and licence, as recorded in the repository's NOTICE file at the moment of use. Every other pretrained model



The shipped file stores every tensor of both networks in half precision and replaces the 400-way Kinetics heads by 40-way heads; the two published checkpoints together would exceed the limit in single precision. Each artefact was weighed on disk by three agreeing counts, and `verify.sh` weighs it again before every run.

Figure 6. The byte budget to scale. The two published checkpoints the networks start from are 172,610,170 B together in single precision. Each submission ships both fine-tuned networks in one fp16 file of 83,543,089 B (022) or 83,559,635 B (026), 16.4 MB under the 100,000,000 B limit that the organizers apply to everything loaded at inference.

named in NOTICE was used only as a probe or as a distillation teacher during development and is not part of either final. Our own code is licensed under Apache-2.0. The ir-CSN architecture is reimplemented in `src/csn.py` from `mmaction2`’s definition so that the published checkpoint loads with strict key matching and no remapping; a mismatch raises an error rather than loading silently.

Table 6. Pretrained weights in the shipped artefacts, and the runtime they need.

Component	Source	Licence
ir-CSN-R50, IG-65M → Kinetics-400 (<code>ircsn_ig65m-pretrained-r50-bnfrozen_8xb12-32x2x1-58e_kinetics400-rgb_20220811-44395bae.pth</code>)	<code>mmaction2</code> model zoo, download. <code>openmmlab.com</code>	Apache-2.0 (<code>mmaction2</code>), over Apache-2.0 (the CSN authors’ release)
ir-CSN-152, IG-65M → Kinetics-400 (<code>ircsn_ig65m-pretrained-r152-bnfrozen_8xb12-32x2x1-58e_kinetics400-rgb_20220811-7d1dacde.pth</code> ; SHA-256 begins <code>a02af4a3a1258bb8</code>)	<code>mmaction2</code> model zoo, download. <code>openmmlab.com</code>	Apache-2.0 (<code>mmaction2</code>)
Runtime: torch 2.13.0, torchvision 0.28.0, numpy 2.4.4, Pillow 12.2.0	PyPI wheels	as recorded in NOTICE (torch: BSD-3-Clause)
Our code (training, inference, packaging)	the submitted repository	Apache-2.0

8 Compliance and disclosures

The statements below follow the order of our signed honor declaration and apply to both final submissions.

1. **Authorship.** The training code, the inference code and the packaging tools were written by this team. Architectures reimplemented from published sources are named in NOTICE with their licences; the ir-CSN backbones are reimplemented from `mmaction2` (Apache-2.0). AI coding assistants (Claude Code) were used during development to help write code, which the

Table 7. Checksums and byte counts of the two artefacts and their submitted CSVs.

File	SHA-256	Bytes
022 artefact, placed as checkpoints/model.pth	218266cc3230e87f441d01abe657e4b2 cf3c87c4294005082114c96c6ef071a9	83,543,089
026 artefact, placed as checkpoints/model.pth	cd27bc72c09f6e4173cd8cf1026e2f96 6eea1bb29ba7288d72e9ea4c3c255940	83,559,635
submissions/022_r152_fence.csv	e9a7c892f9ce62edea4573fbcbeff735 bc1262211b10e95cc05b176a0cfad7ff	16,097
submissions/026_hr320_ person_pixels.csv	50a31f49b04fdb85d1b08c45472dbea2 68f16b870b75755c8d2e7681c37f5801	16,099

host permits in writing (Kaggle discussion 734967: coding assistants “may be used as coding assistants during development”, the rule restricting “the submitted model itself”). No such tool is part of the solution and none is called at inference.

2. **Pretrained weights.** Each artefact contains exactly two trained networks, initialized from mmaction2’s IG-65M $\rightarrow$ Kinetics-400 checkpoints and fine-tuned on the competition’s training split; nothing else is loaded at inference. These are small pretrained video CNNs of the kind the host confirmed as permitted in this track (discussion 738333: an IG-65M $\rightarrow$ Kinetics-400 network “counts as a permitted pretrained CNN, not a prohibited large backbone”; discussion 734967: “Small pretrained models: Allowed”).
3. **Data.** No dataset other than the competition’s own training data was used to train either artefact. No test label of any kind was used or inferred. Neither final uses pseudo-labels; an earlier submission (023) did, and it is not a final. No feature, split or analysis was derived from filenames, timestamps or repository metadata.
4. **Not a language model, offline.** The artefacts are convolutional networks. They call no external API and no language model, and they run without network access, which is tested inside a network namespace with the loopback interface down.
5. **Size.** Everything loaded at inference is one file: 83,543,089 B (022) or 83,559,635 B (026), under the 100,000,000 B limit the host applies to the total of all models used at inference, and under our own 95 MB gate, which `verify.sh` asserts.
6. **Reproducibility.** A fresh clone of the repository at the submitted commit plus the one file reproduces the submitted CSV byte for byte through `./inference.sh` on CPU (405 of 405 rows for both artefacts), and a random 40-clip subset reproduces the full run’s predictions clip for clip.
7. **Selection of the finals.** The two final submissions were chosen by rules written before their scores existed, never by the public leaderboard maximum (Section 9).
8. **Record.** Every claim in this document is traceable to a dated row of the team’s decision ledger and to the audit logs archived beside it; we will make the ledger available to the committee on request.

The repository is private during the competition and, should the team place in the top six, will be published under Apache-2.0 within 30 days of the finals as the competition’s IP clause requires.

9 How the two final submissions were chosen

We fixed the selection rules before any candidate’s score existed, and Figure 8 places the decisions in time.

- (i) A final is never selected by its public score alone; a pick needs a reason other than “it scored

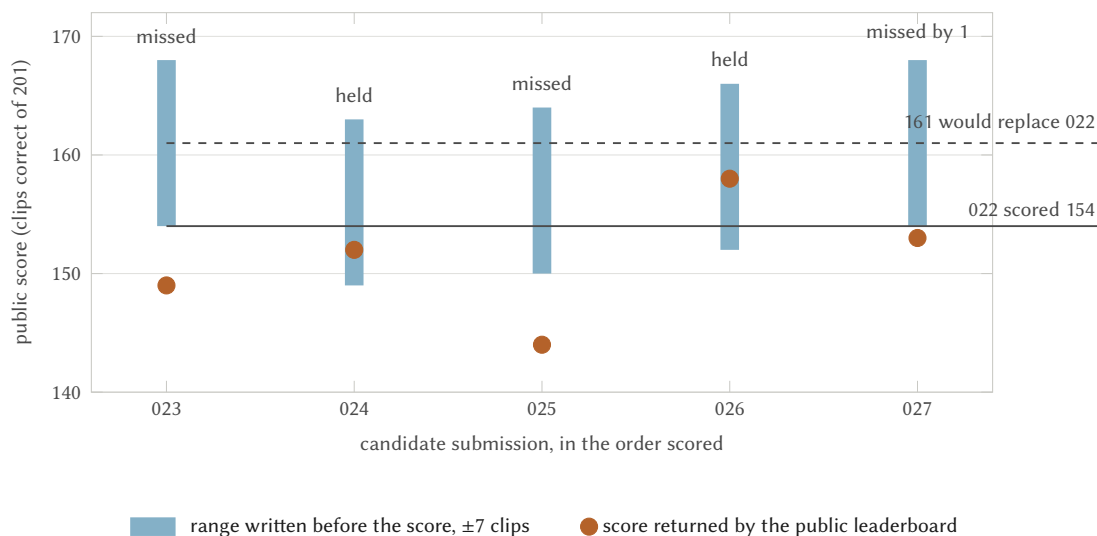


Figure 7. The five candidates scored after 022, each against the public range written into the record before it was submitted. The range is the candidate’s fold result converted at 3.01 clips per fold percentage point, plus or minus the seven-clip noise floor of a 201-clip score. Two ranges held and three were missed. Submission 026 is the only candidate that both passed the fold gate and held its range, and it did so above 022; none reached the 161 that would have replaced 022 as the first pick.

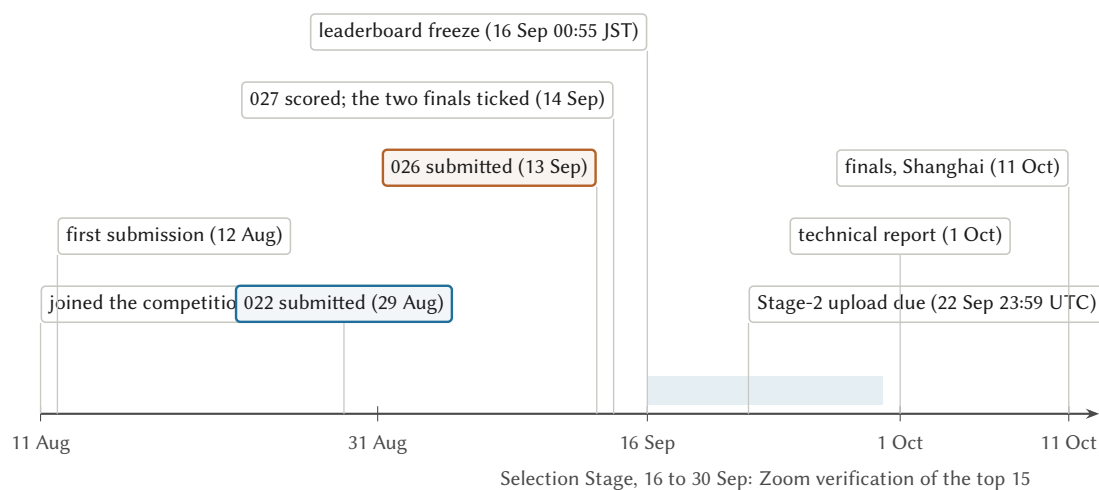


Figure 8. The competition timeline. Submission 022 was made on 29 August and held the public lead of the team’s entries for two weeks; 026 followed on 13 September after its recipe cleared the fold gate. The two finals were selected on 14 September, two days before the leaderboard freeze. If the team places in the private top 15, the Stage-2 upload and the Zoom verification follow within the Selection Stage.

highest”.

- (ii) Submission 022, the incumbent, stays the first pick unless a challenger scores at least 161 of 201, that is, seven clips above 022, one standard error of a public score.
- (iii) The second pick is the candidate with the highest expected accuracy on the private half, judged first by the pre-registered fold instrument of Section D and then by the public reading. The public 201 and the private 204 clips come from the same four test subjects, so a public difference between two of our files is a direct, if noisy, measurement of the private difference. Before each candidate was submitted, its expected public range was written into the record from its fold result and a measured conversion of 3.01 public clips per fold percentage point, with a width of ± 7 clips. Figure 7 shows the five scored candidates against those ranges.

Submission 026 is the second pick on all three counts. Its fold result, $+1.715$ pp with $t = +3.68$ over six subject-disjoint cells, is the largest on the record among recipes trained without any test input; the pseudo-labelled 023 read $+2.080$ pp on the subjects whose test clips it had seen and $+0.506$ pp on held-out subjects. Submission 026 is also the only one of the four fold-passing candidates whose written range held: 158 against a range of 152 to 166, where the pseudo-labelled 023 scored 149 against 154 to 168, the ViT-B container 025 scored 144 against 150 to 164, and 027 scored 153 against 154 to 168. Its per-subject reading resembled the one earlier change that had carried to the board (15 of 18 held-out subjects improved, the worst by -1.97 pp). Its compliance class is 022’s, and its installation from a clean clone was rehearsed before the picks. On the public reading, 026 is four clips above 022 on the 42 clips where the two files differ; our own estimate of the probability that 026 also scores above 022 on the private half is about 0.73, a planning estimate and not a measurement.

Submission 022 stays the first pick because no candidate reached 161, and because it is the better partner for 026: the two files disagree on 42 of 405 clips, so selecting both covers the case in which the thermal change does not carry to the private half. The other scored candidates are not finals. Submission 027 applied the same native-resolution change to the depth and infrared network as well; it passed the fold gate but its per-subject shape failed our subject-level check, and it scored one clip below its range and five below 026. Submission 024 is 022’s own weights with a different rule for combining the two branches; it scored two clips below 022. Submission 025 replaced the thermal network with a ViT-B and scored ten clips below 022, six below its range. Submission 023 was trained with pseudo-labels of test inputs and scored five below 022. The two ticks were made on 14 September 2026 at 09:48 JST, two days before the freeze.

10 Limitations

The first limitation concerns the evidence for the change in 026. On the fold instrument the thermal change is $+1.715$ pp with $t = +3.68$ over six subject-disjoint cells; on the public leaderboard it is four clips of 201, inside one standard error. Of the four changes that passed the same fold gate and were then scored, three did not hold their written range and one, this one, did. We therefore describe 026 as consistent with its fold result and not as a measured gain, and we selected 022 beside it for that reason.

The second limitation concerns the spread between subjects. The same recipe scores between 62.41% and 83.69% across the 18 training subjects when each is held out, a range of 21.28 pp. A verification sample of a few unseen subjects therefore carries a sampling term of its own, about 4.2 pp of standard deviation for two subjects, before any property of the model is measured.

The third limitation concerns the seen half of the verification sample. Every shipped member was trained on all 18 training subjects, so clips of those subjects are in-sample, and their accuracy is not comparable to the held-out figures in this document.

The fourth limitation concerns class support. The training split is imbalanced 30.4 to 1 between its most and least frequent class; the least frequent, *Watch TV*, has 12 clips from 3 subjects, and our fold instrument has no validation clip of that class in fold 1. Both finals nevertheless predict all 40 classes on the test set.

The fifth limitation concerns cost. The thermal network of 026 needs 1.8 times the CPU time of 022's per clip (3.45 s against 1.91 s) and 1.4 times the GPU time. Both artefacts finish the 405 test clips well inside the two-hour ceiling of the live verification run.

References

- [1] D. Tran, H. Wang, L. Torresani, and M. Feiszli. Video classification with channel-separated convolutional networks. In *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [2] D. Ghadiyaram, D. Tran, and D. Mahajan. Large-scale weakly-supervised pre-training for video action recognition. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [3] W. Kay et al. The Kinetics human action video dataset. *arXiv:1705.06950*, 2017.
- [4] MMAAction2 Contributors. OpenMMLab's next generation video understanding toolbox and benchmark. <https://github.com/open-mmlab/mmaaction2>, 2020.
- [5] L. Wang, Y. Xiong, Z. Wang, Y. Qiao, D. Lin, X. Tang, and L. Van Gool. Temporal segment networks: towards good practices for deep action recognition. In *Proc. European Conference on Computer Vision (ECCV)*, 2016.
- [6] P. Izmailov, D. Podoprikin, T. Garipov, D. Vetrov, and A. G. Wilson. Averaging weights leads to wider optima and better generalization. In *Proc. Conference on Uncertainty in Artificial Intelligence (UAI)*, 2018.
- [7] J. Cha, S. Chun, K. Lee, H.-C. Cho, S. Park, Y. Lee, and S. Park. SWAD: domain generalization by seeking flat minima. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

A The 42 clips on which the two submissions differ

Table 8 lists every test clip for which 022_r152_fence.csv and 026_hr320_person_pixels.csv carry different predictions, counted from the two submitted files. Clips are the competition's SM_test_<number> directories, given by number; classes are given as their index and name.

B Held-out accuracy per training subject

Table 9 gives the 022 recipe's accuracy on each training subject when that subject is held out, the seed mean of two members trained on the other 12 subjects of its fold. The fold means re-derived from these rows reproduce the cells of Table 4 to three decimals.

C To be filled on the day of verification

The private leaderboard decides which artefact is reproduced. The following are read aloud and recorded at the start of the session.

Item	Value
The reproduced submission	022 / 026
Its private score	
Commit hash of the submitted repository	
sha256sum checkpoints/model.pth	must equal the row of Table 7
CSV compared by verify.sh	submissions/022_r152_fence.csv / submissions/026_hr320_person_pixels.csv

Table 8. The 42 test clips with different predictions in 022 and 026.

Clip	022	026	Clip	022	026
0020	17 Tap the keyboard	10 Stir drinks	0222	8 Take and use tableware	9 Pour drinks
0023	11 Peel fruits	14 Wipe bowls	0227	11 Peel fruits	22 Turn pages
0024	26 Play games	18 Write	0235	26 Play games	20 Check the time
0029	6 Drink water	7 Eat food	0246	20 Check the time	21 Read documents
0036	37 Take medicine	39 Take body temperature	0254	10 Stir drinks	9 Pour drinks
0038	11 Peel fruits	22 Turn pages	0265	24 Use a mobile phone	26 Play games
0044	9 Pour drinks	8 Take and use tableware	0273	11 Peel fruits	10 Stir drinks
0047	8 Take and use tableware	24 Use a mobile phone	0284	8 Take and use tableware	18 Write
0075	37 Take medicine	6 Drink water	0294	8 Take and use tableware	31 Do stretching exercises
0085	8 Take and use tableware	39 Take body temperature	0299	7 Eat food	38 Massage oneself
0091	26 Play games	11 Peel fruits	0301	37 Take medicine	39 Take body temperature
0100	19 Make a phone call	34 Sit down	0337	11 Peel fruits	1 Brush teeth
0105	21 Read documents	18 Write	0340	7 Eat food	38 Massage oneself
0116	18 Write	26 Play games	0342	24 Use a mobile phone	7 Eat food
0121	9 Pour drinks	10 Stir drinks	0360	8 Take and use tableware	39 Take body temperature
0138	20 Check the time	10 Stir drinks	0364	13 Mop the floor	12 Sweep the floor
0180	26 Play games	33 Lie down	0368	10 Stir drinks	21 Read documents
0198	11 Peel fruits	4 Wipe hands	0372	21 Read documents	17 Tap the keyboard
0201	21 Read documents	22 Turn pages	0376	11 Peel fruits	8 Take and use tableware
0218	10 Stir drinks	22 Turn pages	0390	10 Stir drinks	8 Take and use tableware
0219	7 Eat food	39 Take body temperature	0404	18 Write	37 Take medicine

Table 9. Held-out accuracy of the 022 recipe per training subject, in percent.

Fold	Subject	Clips	Seed 1	Seed 2	Mean
0	4	143	72.73	72.73	72.73
0	5	160	78.12	81.25	79.69
0	9	187	83.42	83.96	83.69
0	21	133	62.41	62.41	62.41
0	22	194	70.10	70.62	70.36
0	24	159	74.21	75.47	74.84
1	2	168	75.60	77.38	76.49
1	6	203	64.04	63.55	63.79
1	8	169	66.86	64.50	65.68
1	17	166	78.92	80.12	79.52
1	18	179	70.95	65.92	68.44
1	19	188	80.85	82.98	81.91
2	1	153	67.97	70.59	69.28
2	3	163	69.94	65.03	67.48
2	7	185	74.59	72.43	73.51
2	16	186	74.19	76.88	75.54
2	20	159	74.84	72.33	73.58
2	23	141	68.09	65.25	66.67